INCH-POUND
MIL-DTL-43021F
30 JUNE 2003
SUPERSEDING
MIL-B-43021E
11 JUNE 1992

DETAIL SPECIFICATION

BINDER, LOAD, CARGO TIE-DOWN, 10,000 POUND CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size load binder of 10,000 pound capacity.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Natick Soldier Center ATTN: AMSSB-RAD-AD(N), Natick, MA 01760-5017, using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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DEPARTMENT OF DEFENSE

MIL-STD-130 - Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 40, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NATICK SOLDIER CENTER

11-1-901 - Binder, Load, Cargo Tiedown, 10,000 pound Capacity

(Copies are available from the Natick Soldier Center, ATTN: AMSSB-RAD-AD(N) – CONFIGURATION MANAGER, Natick, MA 01760-5017.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to a first article inspection in accordance with 4.2.

3.2 Materials. Materials shall conform to the requirements of the Technical Data Package (TDP) to subsidiary specifications and standards applicable thereto, and as specified herein.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the materials meet or exceed the operational requirements, and promote economically advantageous life cycle costs.

3.3 Construction. Details of construction shall be in accordance with the TDP.

3.4 Performance characteristics.

3.4.1 Proof load. The assembled load binder during the proof load test (see 4.6.2) shall show no evidence of a release failure. After testing, the release shall be disassembled and the

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components examined. Components shall have no visible evidence of a fracture, weld separation or deformation.

3.4.2 Ultimate load. The assembled load binder during the ultimate load test (see 4.6.3) shall show no evidence of rupture. Deformation is allowed.

3.5 Identification marking. Each load binder shall be marked for identification with the capacity, 10,000 LB, manufacturer's name or readily identifiable trade mark, and the Part Number 11-1-901. Part numbering shall be in accordance with MIL-STD-130. The markings shall be permanently indented, located as shown on the drawing, and of such size as to be clearly legible.

3.6 Workmanship. All components and assemblies of the load binder assembly shall be free from dirt and other extraneous material, burrs, slivers, dents, cracks, fractures and damaged welds. External surfaces shall be free from sharp edges and corners. Components shall not be buckled, loose, dented, missing, malformed or misaligned. The load binders shall conform to the quality of product established by this specification.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete load binder assembly, unless otherwise stated in the contract, when a first article sample is specified (see 3.1). The inspection shall include the tests and examinations of 4.3.1, 4.3.2, 4.4, 4.5 and 4.6. Any defect or nonconformance shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspections shall include the examinations of 4.3.1, 4.3.2, 4.4, 4.5 and 4.6. Sampling plans and acceptance/rejection criteria for 4.4, 4.5 and 4.6 shall be as specified in the contract or purchase order. Binder assemblies which fail the acceptance criteria shall be cause for rejection of the lot.

4.3.1 Component and material inspection. Components and materials shall be inspected in accordance with all of the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details and dimensional requirements, which cannot be examined in the finished product, are in accordance with specified requirements. The Government reserves the right to exclude from consideration of acceptance any material or service for which in-process inspection has indicated nonconformance.

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4.4 End item visual examination. The end items shall be examined for the defects listed in table I. Any nonconformance to the TDP shall be classified as a defect.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish, protective	Any breaks through coating of metal		201
	Any area of rust or corrosion	101	
	Dirt, oil, grease, or other foreign matter		202
Construction and workmanship	Any component missing	102	
	Any component cracked, fractured or otherwise damaged, including damaged welds	103	
	Any sharp edge, burr or metal sliver	104	
	Any part not assembled or joined as specified	105	
Identification Marking	Missing, incomplete not legible, incorrect or not located or applied as specified		203

4.5 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on Drawing 11-1-901. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect.

4.6 Methods of inspection.

4.6.1 End item testing. End item testing for first article inspection shall be in accordance with 4.6.1 and 4.6.2 for compliance with 3.4.1 and 3.4.2. The first article inspection shall be destructive in nature and the sample shall not form part of a production lot. For conformance examinations, only the proof load test (see 4.6.2) shall be performed. The proof load test is nondestructive. Samples, after successful completion of the proof load test, shall form part of the production lot. Results of proof load and ultimate load testing shall be reported as pass or fail.

4.6.2 Proof load test. The binder shall be secured in a suitable tensile testing device and subjected to a dynamic load of 10,000 lbs gradually applied and held for 30 seconds. The direction of the load shall be as depicted on drawing 11-1-901. During the proof load test the binder shall be observed for displaying any evidence of release. After load relaxation, the binder shall be immediately examined for noncompliance to the requirements of 3.4.1. Any noncompliance to 3.4.1 or showing evidence of release during the proof load test shall constitute a failure.

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4.6.3 Ultimate load test. The binder shall be secured in a suitable tensile testing device and subjected to a dynamic load of 13,000 lbs gradually applied and held for 30 seconds. The direction of the load shall be as depicted in drawing 11-1-901. Any noncompliance to 3.4.1 shall constitute a failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The load binder is designed for use as a tie-down tensioning device in rigging general cargo and equipment for transport and air delivery.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of DoDISS to be cited in the solicitation and if required, the specification of individual documents referenced (see 2.2.1).
c. When a first article is required (see 3.1).
d. Sampling plan (see 4.3).
e. Packaging requirements (see 5.1).

6.3 Subject term (keyword listing).

Air delivery
Rigging
Tensioning device

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6.4 Changes from previous issue. Marginal notations are not used in this version to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL2
Air Force - 99

Review Activities:
Air Force - 11

Preparing Activity:
Army - GL2

(Project 1670-1045)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-43021F	**2. DOCUMENT DATE (YYYYMMDD)** 20030630

3. DOCUMENT TITLE

Binder, Load, Cargo Tiedown, 10,000 Pound Capacity

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *Include Area Code)*
Natick Soldier Center	(508) 233-4258
c. ADDRESS *(Include Zip Code)* AMSSB-RAD-AD(N), J. Riley Natick, Ma 01760-5017	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**

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